Does Site Invariance Help at Unseen Hospitals? A Frozen Leave-Site-Out and Conformal Evaluation on Chest Radiographs

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***Abstract*—Diagnostic image models often perform poorly at hospitals absent from their training data. We tried to build a scanner- and institution-invariant pipeline from a class-conditional site adversary, augmentation based on measured per-site image statistics, and conformal prediction sets meant to hold at an unseen site. We tested it on an open chest-radiograph benchmark (924 images, 10 sources, COVID-19 versus other findings), holding out one site at a time. Four sites guided all design decisions; the other four were run once with settings frozen beforehand. The invariance ideas did not help: over eight held-out sites, none of eight alternative training methods beat plain empirical risk minimization (ERM) by more than noise (largest gain $+0.015$ AUROC, GroupDRO; every 95% interval included zero), and the result repeated with a frozen pretrained chest-radiograph encoder (largest gain $+0.018$). The useful findings concern evaluation and uncertainty. Five acquisition statistics predicted the diagnosis on unseen sites (AUROC 0.662). Validation on seen sites overstated held-out AUROC by 0.118, and standard conformal sets covered only 0.836 of cases at a 0.90 target (0.834 with the pretrained encoder). Leave-site-out CV+ calibration removed the worst shortfalls with sets about 26% larger; all calibrations followed one coverage–size curve. Averaging a from-scratch CNN with the pretrained model gave the best mean and worst-site AUROC (0.780 and 0.671; $+0.053$ [-0.013, $+0.129$] over one CNN), consistent across fold sets but not significant. The data are small and cover one modality; we make no claim about clinical COVID-19 detection.**

***Index Terms*—Chest radiography, conformal prediction, dataset shift, domain generalization, leave-one-site-out evaluation, negative results, uncertainty quantification.**

I. Introduction

A DIAGNOSTIC model trained on pooled data from several hospitals can learn cues that identify the hospital instead of the disease [1]–[3]. It then looks good on validation data from the same hospitals and fails quietly at a new one. A common response is to make the model invariant to the scanner and institution through adversarial alignment, feature-statistics matching, harmonization, or augmentation [4]–[7].

We started this project to build such a recipe. It had three parts. First, a site adversary that works within each class, so that different disease prevalence across hospitals is not penalized [8]. Second, augmentation that re-renders an image as if it came from another *observed* site, using statistics estimated from the training images without labels. Third, conformal prediction sets calibrated with an unseen site in mind. We then ran what we thought would be the deciding experiment: leave-one-site-out (LOSO) testing with a development/confirmation split and configurations frozen before the confirmation run.

The recipe did not work. Neither our method nor the baseline methods were reliably better than ERM. A natural objection is that the encoder was a small network trained from scratch, so we repeated the study on top of a frozen pretrained chest-radiograph encoder; the answer did not change. What survived are results about *why* and about reporting uncertainty honestly for sites the model has not seen. We report both, since a frozen negative result is useful for anyone choosing among these methods on small multi-site data. Our contributions are modest, and we carefully mark novelty (our literature search was targeted, not systematic):

- a frozen LOSO protocol on open data with a hierarchical paired bootstrap, a negative result for eight methods against ERM (three of them our own variants), the same negative result on a second, pretrained encoder family, and a controlled label-shift sweep;
- two label-free diagnostics of acquisition shortcuts, one measuring leakage of the label and one measuring how much a model relies on it;
- evidence on calibrating prediction sets for unseen sites, replicated on two encoder families and their ensemble: seen-site validation is optimistic, standard conformal sets under-cover, a site-fold CV+ rule avoids the worst cases, and simulated shifts do not reproduce the real effect;
- a pre-specified, suggestive but not significant, improvement from averaging a from-scratch CNN with a pretrained model.

We do not claim clinical COVID-19 detection performance; the benchmark is a shortcut-prone stress test.

II. Related Work

Shortcuts in chest radiography. Pneumonia and COVID-19 classifiers depend on site and dataset artifacts [1], [9]. Masking the lungs barely changes classifier accuracy on public COVID-19 collections [10], classifiers can prefer shortcuts to signal [2], and a large review found bias throughout this literature [11], [12]. Our diagnostics follow this line and add a held-out-site measurement.

Invariance and domain generalization. DANN, conditional adversaries, CORAL, GroupDRO, IRM, and MixStyle target shift [4], [5], [13]–[16]. DomainBed found that, with fair model selection, no method reliably beats ERM [17],

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and WILDS reported similar difficulty on real shifts [18]. Exact invariance is provably limited when label priors differ across domains [8], and methods that handle joint conditional and label shift assume target data [19], [20]. We studied the setting with no target data. Augmentation work (BigAug, stain augmentation, ComBat) is the closest relative of our site-profile augmentation [6], [7], [21].

Pretrained encoders and ensembles. Pretraining on natural images gave limited gains for several medical imaging tasks [22], while chest-radiograph encoders trained on many public collections are available as a library [23]. Averaging independently trained networks is a simple way to improve accuracy and uncertainty estimates [24]. We use both ideas as a second test of the same question and do not claim them as new.

Conformal prediction under shift. Split conformal prediction requires exchangeability [25], [26]. Weighted and non-exchangeable versions handle known covariate shifts, drift, label shifts, and several sources [27]–[30]. For unseen domains, worst-case calibration over source domains or divergence balls exists [31], and per-domain robust calibration is a baseline in set-valued domain generalization [32]; coverage failures of conformal sets under domain shift in medical imaging have been documented [33]. Our worst-site mode is a variant of these and not new. CV+ comes from jackknife+ [34]. We did not find an evaluation of CV+ with sites as folds on real multi-site medical images, but we did not search exhaustively.

III. Methods

A. Setting and training methods

Data come from K source sites that differ in image acquisition $P_k(X \mid Y)$ and prevalence $P_k(Y)$. Models are trained, selected, and calibrated on the source sites and then tested on one held-out site. The task is binary (COVID-19 vs. other). We denote z as the encoder output and s as the site index.

All methods share one encoder, optimizer, and budget. Baselines: ERM, GroupDRO (sites as groups), DANN (unconditional site adversary, weight 0.5), and generic augmentation (intensity, gamma, noise, blur, and bias-field changes applied per sample). The variants we implemented:

- *Class-conditional adversary.* One site discriminator per class behind a gradient-reversal layer, with each sample sent to the discriminator of its label: $\mathcal{L}_{\text{adv}} = \mathbb{E}\,\mathrm{CE}(d_y(\mathrm{GRL}_\lambda(z)), s)$. A discriminator that only sees one class cannot exploit the prevalence differences. We pair it with per-class CORAL and site-prior reweighting of the task loss, $w_{s,y} = \pi(y)/\pi_s(y)$.
- *Site-profile augmentation.* We measure five statistics per image (mean, contrast, noise floor, sharpness, and low-frequency bias), summarize them per training site without labels, and translate an image from its site toward another observed site by applying only the relative change. It is a measurement-driven proxy, not a scanner simulator.
- *Profile adversarial invariance (PAI).* A gradient-reversal head attempts to regress the measured profile of the input from z.
- *SIDL variants.* SIDL-cc: generic augmentation, class-conditional adversary, per-class CORAL, prior reweighting, and consistency loss between clean and augmented views. SIDL-phys: site-profile augmentation, unconditional adversary, pooled CORAL, consistency. SIDL-full: site-profile augmentation plus SIDL-cc components.

Mild settings (augmentation probability 0.3, strength 0.5, adversary weight 0.1 for SIDL and PAI) were chosen on the development folds and frozen.

B. Second encoder family: frozen pretrained features

To test whether the conclusions depend on a small from-scratch CNN, we froze the DenseNet-121 [35] chest-radiograph encoder from the TorchXRayVision library [23] (the "all" weights, trained on several public radiograph collections without COVID-19 labels), extracted its 1024-dimensional embedding of each 224×224 image once, and trained only a small head: a linear layer to 256 units with ReLU and dropout 0.2, followed by a linear classifier (30 epochs, AdamW, learning rate 10^{-3}, batch size 64, best source-validation AUROC). The head was trained with the same methods as above (ERM, prior reweighting, GroupDRO, DANN, class-conditional adversary with and without prior reweighting, CORAL, and the class-conditional recipe without image augmentation) at their default weights (0.5 adversary, 0.1 CORAL), plus per-site feature centring. Centring subtracts a site mean from the features and uses the unlabeled target images for the held-out site, so it is transductive and not comparable with the other heads. Features are standardized with source-site statistics only. An *ensemble* is the mean predicted probability of the models named; the combined model averages five CNNs and five pretrained heads (seeds 0–4 each).

C. Diagnostics

Leak: a logistic regression on the five profile numbers is trained on source sites and scored on the held-out site; an AUROC above 0.5 means that acquisition statistics alone, without anatomy, predict the label. *Reliance:* within each class, the Spearman correlation between a model's held-out $P(\text{COVID-19})$ and that profile-only score, averaged over classes; working within class means that the label cannot explain the correlation.

D. Conformal calibration

We use the score $1 - p_\theta(y \mid x)$ [36] and sets $\{y : 1 - p_\theta(y \mid x) \le \hat{q}\}$. Three standard variants use the validation split of seen source sites: pooled, site-balanced (equal weight per site), and worst-site (largest per-site quantile among sites with at least 25 images). For the leave-site-out (LSO) variants, we train an inner model for every source site k on the other source sites and score *all* images of site k with it, giving scores R_i that mimic behavior on an unseen site. LSO-split applies the pooled quantile of the R_i to the deployed model. LSO-worst takes the largest per-site quantile. LSO-CV+ applies the CV+ rule with sites as folds: label y enters the set for test image x iff $\#\{i :$

TABLE I
BENCHMARK COMPOSITION. ROLE IS THE FOLD ROLE WHEN THE SOURCE IS HELD OUT; FIGURE1 AND HANNOVER-DE HAVE TOO FEW NEGATIVES AND ARE ONLY USED FOR TRAINING.

Source	Images	COVID-19	Other	Prevalence	Role
Actualmed	185	58	127	0.31	development
China	56	51	5	0.91	development
Italy	120	100	20	0.83	development
Spain	56	38	18	0.68	development
Australia	60	6	54	0.10	confirmation
Other	207	122	85	0.59	confirmation
UK	66	49	17	0.74	confirmation
Unknown	52	33	19	0.63	confirmation
Figure1	40	35	5	0.88	training only
Hannover-DE	82	79	3	0.96	training only
Total	924	571	353	0.62	

$R_i < 1 - p_{-k(i)}(y \mid x)\} < (1-\alpha)(n+1)$, where every inner model scores x. CV+ guarantees coverage for exchangeable *points* [34]; here, the exchangeable units would be sites, we have only nine, and we have no theorem. Coverage is measured empirically. Besides coverage and mean set size at $\alpha = 0.1$, we report the coverage-calibration error (mean $|\text{coverage} - (1-\alpha)|$ over nine values of α from 0.04 to 0.30), the fraction of (site, α) cells more than five points under target, and set size interpolated at 90% coverage.

IV. BENCHMARK AND PROTOCOL

A. Data

We pooled three public GitHub collections of frontal chest radiographs [37]–[39]. A site is the institution or country in the metadata. Small sites are merged into "Other" and images with no location form "Unknown", so these two groups mix institutions. The label is COVID-19 versus any other finding. Images are intensity-normalized, converted to grayscale, and area-downsampled to 64×64 for the CNN (and to 224×224 for the pretrained encoder). The benchmark has 924 images from ten sources (Table I), and prevalence varies a lot between them (SD 0.274); therefore, label shift matters.

B. Folds and freezing

Eight sources were each held out once; Figure1 and Hannover-DE, which have only 5 and 3 negative images, were used for training only. We assigned four to development (Actualmed, Italy, China, and Spain) and four to confirmation (UK, Australia, Other, and Unknown) before looking at any confirmation result; the split was not randomized. Every design decision used development folds only. At the end of iteration 8, we froze nine CNN configurations and ran them once on the confirmation folds. Validation (a patient-level 15% split) and checkpoint choice (best source-validation AUROC) never touched the held-out site. One exception is descriptive: the label-free diagnostics were computed once over all eight sites in the first iteration; they involved no tuned model and no choice among methods used them.

The pretrained family was added afterward (iterations 11–17 in the released log) under the same rules. Its head settings were examined on the development folds only (seven settings gave a mean AUROC of 0.737–0.743, so the defaults were kept). Before its first confirmation run we wrote down the settings of all nine heads and two primary comparisons: (A) the combined ensemble against the five-CNN ensemble, and (B) the same ensemble against a single CNN. Then we ran the confirmation folds once. Two disclosures follow. The CNN confirmation results were already known when the pretrained family was designed, and three further CNN ERM seeds were run after the freeze with the unchanged configuration so that both families have five seeds.

C. Models and statistics

The CNN encoder has four conv–BatchNorm–ReLU–max-pool blocks (widths 16–64), trained from scratch with AdamW (learning rate 2×10^{-3}), batch size 64, 15 epochs, and two seeds for the nine-method comparison (five seeds for ERM in the family comparison), on two CPU cores (25–80 s per run). The unit of generalization is the held-out site. For each method versus ERM, we take the paired AUROC difference (seed-averaged) per site, then bootstrap hierarchically: resample sites, then resample test images within each chosen site stratified by class, using the same images for both methods (1500 replicates; 2000 for the family comparison). This is wider than a bootstrap over sites alone, which matters because some sites are tiny (Australia has 6 COVID-19 images; China has 5 negatives). For the label-shift and acquisition-shift experiments, we also build a semi-synthetic benchmark: images are assigned at random to five virtual sites with known transforms (gamma, contrast, blur, noise, and bias field) and controlled prior spread. Because the family comparison averages five CNN seeds, the CNN ERM mean over eight sites is 0.727 there and 0.733 (two seeds) in Table II.

V. RESULTS

A. Acquisition statistics leak labels

Five numbers with no anatomy predicted the diagnosis on held-out sites with a mean AUROC of 0.662 (gradient boosting: 0.588; Fig. 1) and predicted the source site with an accuracy of 0.367 against 0.224 for guessing the majority site. Re-measuring the leak on site-profile-augmented training images did not reduce it ($0.662 \rightarrow 0.660$); therefore, the label-correlated signal seems to live mostly *within* sites (for example, portable versus fixed acquisition or patient severity) and not in site-level signatures that augmentation can randomize. Model scores track the same signal: within-class reliance is 0.438 for ERM and between 0.396 and 0.514 for the other methods. Site-profile augmentation and PAI lowered it only slightly (0.396, 0.398), and the two methods with a class-conditional adversary and prior reweighting had the highest values (0.473, 0.514).

B. No method significantly beats ERM

ERM reached AUROC of 0.758 on development folds and 0.707 on confirmation folds (Table II, Fig. 2). Paired differences to ERM are small, and all intervals include zero: GroupDRO $+0.015$ $[-0.015, 0.055]$, SIDL-phys $+0.011$

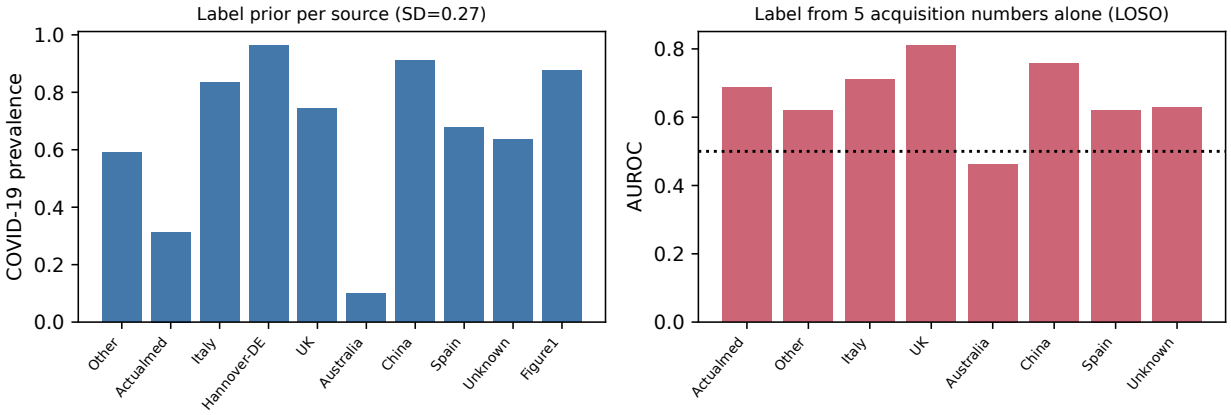


Fig. 1. Left: COVID-19 prevalence per data source. Right: AUROC for predicting the diagnosis on a held-out site from five acquisition statistics alone.

TABLE II
LOSO TASK METRICS FOR THE CNN (MEAN OVER HELD-OUT SITES OF THE TWO-SEED MEAN). Δ IS THE PAIRED AUROC DIFFERENCE TO ERM OVER EIGHT SITES WITH A HIERARCHICAL BOOTSTRAP 95% INTERVAL. "WORST FOLD" IS THE LOWEST PER-SITE AUROC. BALANCED ACCURACY AND ECE USE THE ARGMAX DECISION.

Method	AUROC dev	AUROC conf.	AUROC all	Worst fold	Δ vs ERM [95% CI]	Bal. acc.	ECE
ERM	0.758	0.707	0.733	0.565	–	0.634	0.157
GroupDRO	0.761	0.735	0.748	0.657	+0.015 [-0.015, +0.055]	0.689	0.144
DANN	0.752	0.702	0.727	0.645	-0.006 [-0.049, +0.035]	0.599	0.166
Generic aug.	0.748	0.716	0.732	0.619	-0.001 [-0.029, +0.032]	0.663	0.150
Site-profile aug.	0.757	0.694	0.726	0.641	-0.007 [-0.060, +0.038]	0.618	0.174
Profile adversary (PAI)	0.753	0.708	0.730	0.551	-0.002 [-0.020, +0.014]	0.615	0.154
SIDL-cc	0.722	0.699	0.711	0.635	-0.022 [-0.069, +0.030]	0.612	0.178
SIDL-phys	0.756	0.732	0.744	0.650	+0.011 [-0.016, +0.050]	0.656	0.192
SIDL-full	0.738	0.665	0.701	0.415	-0.031 [-0.085, +0.016]	0.590	0.175

$[-0.016, 0.050]$, generic augmentation -0.001, PAI -0.002, DANN -0.006, site-profile augmentation -0.007, SIDL-cc -0.022, SIDL-full -0.031 $[-0.085, 0.016]$. Upper bounds rule out gains above about 0.055 for every method. Orderings are unstable: method means correlate only 0.500 (Spearman) between development and confirmation folds. Site-profile augmentation tied ERM on development folds (0.757 vs. 0.758) and fell below it on confirmation folds (0.694 vs. 0.707); SIDL-phys moved the other way. GroupDRO also has higher balanced accuracy (0.689 vs. 0.634) and a better worst-site AUROC (0.657 vs. 0.565), but ERM's worst site has only 6 positives, PAI does not share the gain, and balanced accuracy depends on an operating point that group reweighting shifts, so we treat both as exploratory. In an early iteration, whole-batch full-strength augmentation cost up to 0.083 AUROC on development folds (0.675 vs. 0.758); making it per-sample and mild removed the drop but never produced a gain.

C. Controlled label shift

On the semi-synthetic benchmark (identical image content, known transforms), DANN, cDANN, and cDANN with prior reweighting all lose to ERM at the largest prior spread (0.4): -0.092 $[-0.175, -0.016]$, -0.077 $[-0.142, -0.019]$, and -0.047 $[-0.092, -0.013]$; reweighting alone is neutral (-0.000). The ordering matches theory (unconditional alignment hurts most, class-conditional with reweighting least), but at weight 0.5, the adversaries already lose 0.02–0.06 AUROC with no label shift at all. Lowering the weight moves them toward ERM and never above it, and the sign of the advantage of cDANN over DANN flips across weights; with six units per cell, we cannot claim that the class-conditional version helps (Table III).

TABLE III
SEMI-SYNTHETIC LABEL-SHIFT SWEEP: HELD-OUT AUROC (VIRTUAL SITES V0, V2, V4, TWO SEEDS, ADVERSARY WEIGHT 0.5).

Prior spread	ERM	ERM+prior	DANN	cDANN	cDANN+prior	Units
0.00	0.732	0.735	0.672	0.682	0.712	6
0.25	0.721	0.725	0.681	0.685	0.681	6
0.40	0.703	0.703	0.611	0.626	0.656	6

D. A frozen pretrained encoder gives the same answer

Replacing the CNN by a frozen chest-radiograph encoder did not change the conclusion. Over eight sites the pretrained head is not clearly better than the CNN (AUROC 0.749 vs. 0.727; difference $+0.022$ $[-0.079, +0.133]$). It is better at five sites (most clearly Australia and China) and worse at Actualmed, Italy, and UK (Fig. 3a), and its mean ranks differently on the two fold sets (0.739 vs. 0.755 on development, 0.758 vs. 0.699 on confirmation). None of the invariance heads beat ERM on the embedding (Table IV). GroupDRO again had the largest gain, $+0.018$ $[-0.000, +0.039]$ ($+0.021$ on development folds, $+0.016$ on confirmation folds), with an interval that touches zero and the status of the best of nine uncorrected comparisons. The class-conditional adversary, CORAL, DANN, and prior reweighting were within 0.006 of ERM; adversary plus prior reweighting was lower (-0.025), and site centring, which needs unlabeled target images, was clearly worse (-0.040 $[-0.076, -0.006]$). Every upper bound was below 0.04. The embedding still carries site information: a patient-grouped linear classifier names the source among ten with accuracy 0.473 (0.224 for the majority source).

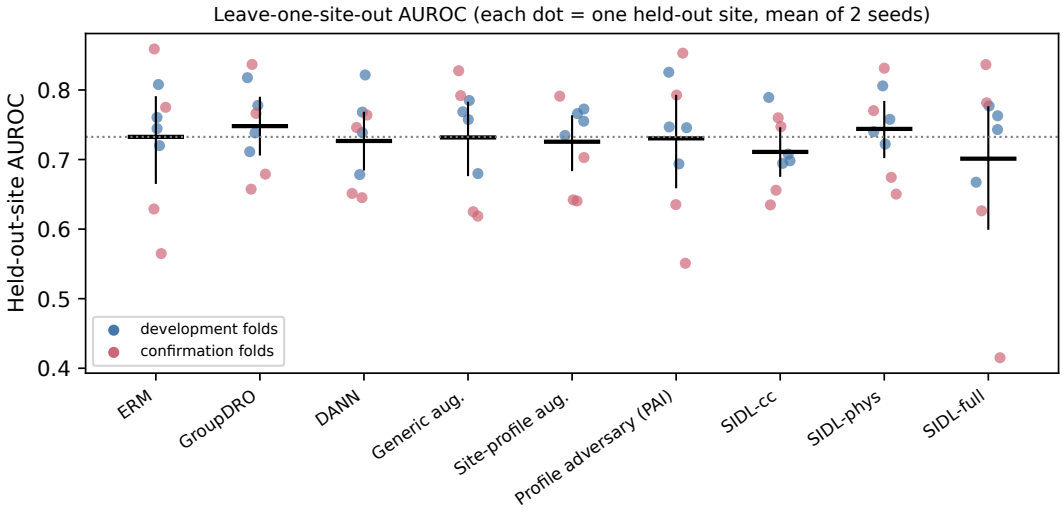


Fig. 2. Held-out AUROC per CNN method (dots: sites; bar and line: mean and bootstrap interval over sites).

TABLE IV
TRAINING METHODS ON THE FROZEN PRETRAINED EMBEDDING (FIVE SEEDS; SETTINGS FIXED BEFORE THE CONFIRMATION RUN). Δ IS THE PAIRED AUROC DIFFERENCE TO THE PRETRAINED ERM HEAD OVER EIGHT SITES WITH A HIERARCHICAL BOOTSTRAP 95% INTERVAL.

Head on frozen embedding	Dev	Conf.	All	Δ vs ERM [95% CI]
ERM	0.739	0.758	0.749	–
GroupDRO	0.760	0.774	0.767	+0.018 [−0.000, +0.039]
Prior reweighting	0.738	0.747	0.742	−0.006 [−0.017, +0.005]
DANN	0.736	0.759	0.747	−0.002 [−0.019, +0.020]
Class-cond. adversary	0.739	0.755	0.747	−0.001 [−0.019, +0.017]
Class-cond. adv. + prior	0.706	0.742	0.724	−0.025 [−0.062, +0.005]
CORAL	0.740	0.755	0.748	−0.001 [−0.012, +0.010]
Class-cond. recipe	0.741	0.743	0.742	−0.006 [−0.026, +0.011]
Site centring (uses target images)	0.715	0.703	0.709	−0.040 [−0.076, −0.006]

TABLE V
ENCODER FAMILIES AND THEIR ENSEMBLES, HELD-OUT AUROC AVERAGED OVER THE FOUR DEVELOPMENT SITES, THE FOUR CONFIRMATION SITES, AND ALL EIGHT (FIVE SEEDS). Δ IS THE PAIRED DIFFERENCE TO THE SINGLE CNN WITH A HIERARCHICAL BOOTSTRAP 95% INTERVAL (2000 REPLICATES).

Model	Dev	Conf.	All	Worst site	Δ vs single CNN [95% CI]
CNN, single (ERM)	0.755	0.699	0.727	0.595	–
Pretrained head, single (ERM)	0.739	0.758	0.749	0.619	+0.022 [−0.079, +0.133]
CNN, 5-model ensemble	0.772	0.716	0.744	0.617	+0.017 [+0.006, +0.030]
Pretrained head, 5-model ensemble	0.750	0.771	0.760	0.646	+0.034 [−0.070, +0.149]
CNN + pretrained head (5+5 ensemble)	0.789	0.770	0.780	0.671	+0.053 [−0.013, +0.129]

E. Combining the two encoder families

The two models fail at different sites. The CNN is weakest on Australia and Other, and the pretrained head on Actualmed and UK. Averaging five CNNs with five pretrained heads gave the best mean and the best worst-site AUROC (Table V, Fig. 3a). The two pre-specified comparisons were positive in direction on both fold sets and not significant: against the five-CNN ensemble $+0.036$ $[-0.031, +0.110]$ (development $+0.017$, confirmation $+0.055$), and against a single CNN $+0.053$ $[-0.013, +0.129]$ (development $+0.034$, confirmation $+0.071$). Two cautions apply. On the confirmation folds the pretrained five-head ensemble was as good as the combination (0.771 vs. 0.770), so the data cannot separate "combining helps" from "pretraining helps". Seed averaging alone also helps: the five-CNN ensemble beats the single CNN by $+0.017$ $[+0.006, +0.030]$. We treat the combination as a suggestive direction and not as a result.

F. Conformal sets under-cover unseen sites

Sets calibrated on seen-site validation data miss the target (Table VI, Fig. 4): mean coverage is 0.836 (pooled) and 0.840 (site-balanced) at the 0.90 target, the worst site gets about 0.743, and 4 of 8 sites are more than five points short. The pattern shows up in both fold sets (0.829 on development, 0.843 on confirmation for pooled). Worst-site calibration restores the mean (0.904) but leaves two sites short (lowest 0.751, at "Other"). LSO-CV+ reaches mean coverage 0.942 and lowest site coverage 0.867 with no site more than five points short, at mean set size 1.747 against 1.389 for pooled. It over-covers on most sites; LSO-split lands nearer the target on average (0.905) but has a site at 0.732. Interpolated at 90% coverage, set size is between 1.574 and 1.603 for *every* calibration (pooled 1.582, LSO-CV+ 1.602). Development folds had suggested that LSO-CV+ was about 4% smaller at matched coverage; the confirmation folds did not repeat that. So the calibrations differ in operating point and not in efficiency, and LSO-CV+'s benefit is safety. Whether sets about 26% larger are worth a six-point average shortfall (up to sixteen at the worst site) depends on the application. Two other ideas failed on development folds and were dropped: weighting calibration scores by a density ratio of the acquisition profile (coverage 0.834 vs. 0.829 for pooled), and correcting for estimated label shift [29], [40] (coverage 0.813 with an EM prior, 0.845 with the oracle prior; balanced accuracy got worse).

The same holds for the pretrained head and the ensemble (Table VII, Fig. 3b). Pooled coverage is 0.834 for the pretrained head and 0.836 for the ensemble, with 5 of 8 sites

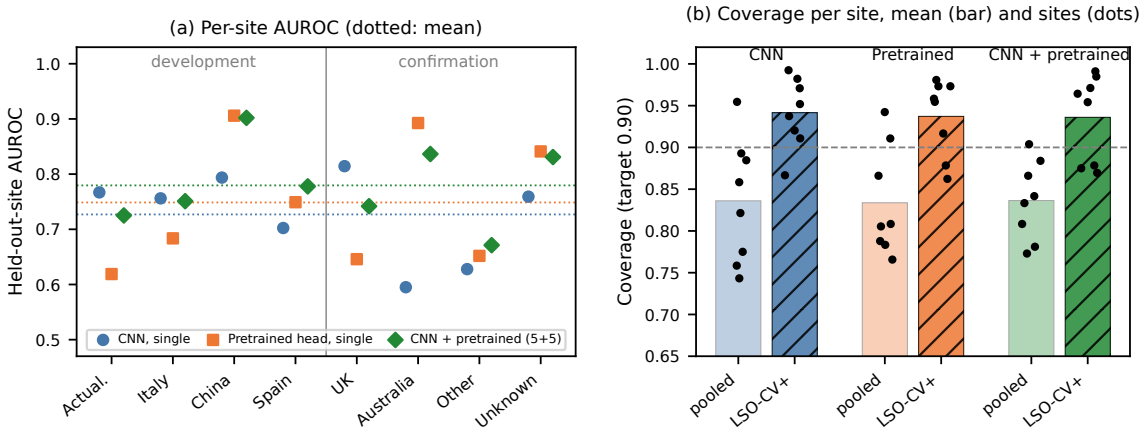


Fig. 3. (a) Held-out AUROC for the single CNN, the single pretrained head, and their combination at each site; dotted lines are means over eight sites. (b) Conformal coverage on each held-out site at $\alpha = 0.1$ for seen-site pooled and LSO-CV+ calibration (bars: mean of eight sites; dots: sites; dashed: target).

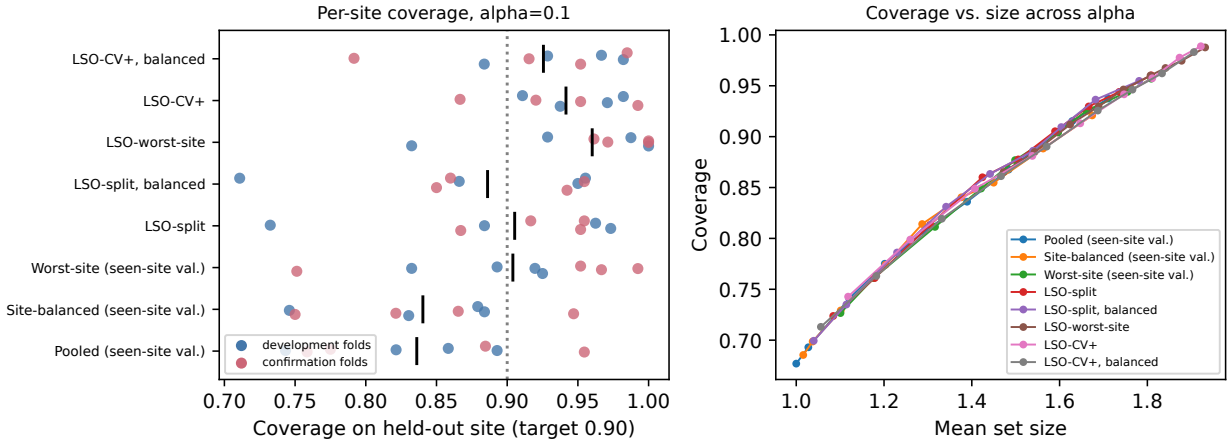


Fig. 4. Left: coverage of each held-out site at $\alpha = 0.1$ (bar: mean). Right: coverage versus set size over the α grid; all calibrations lie on nearly one curve.

TABLE VI
CONFORMAL PREDICTION FOR THE CNN ERM MODEL ON EIGHT HELD-OUT SITES, $\alpha = 0.1$ (TARGET 0.90; TWO SEEDS). "FOLDS <0.85" COUNTS SITES WITH MORE THAN FIVE POINTS UNDER TARGET; THE LAST THREE COLUMNS USE THE α GRID. ROWS 1–3 WERE CALIBRATED ON THE SEEN-SITE VALIDATION DATA, AND ROWS 4–8 USED LEAVE-SITE-OUT SCORES.

Calibration	Cov.	Min cov.	Folds <0.85	Size	Cov. calib. err.	Frac. under	Size@90%
Pooled (seen-site val.)	0.836	0.743	4/8	1.39	0.048	0.389	1.58
Site-balanced (seen-site val.)	0.840	0.746	4/8	1.38	0.053	0.472	1.60
Worst-site (seen-site val.)	0.904	0.751	2/8	1.60	0.009	0.236	1.58
LSO-split	0.905	0.732	1/8	1.59	0.011	0.208	1.57
LSO-split, balanced	0.886	0.711	1/8	1.54	0.009	0.236	1.58
LSO-worst-site	0.960	0.832	1/8	1.81	0.083	0.028	1.59
LSO-CV+	0.942	0.867	0/8	1.75	0.041	0.069	1.60
LSO-CV+, balanced	0.926	0.792	1/8	1.69	0.020	0.153	1.60

more than five points short in both, and LSO-CV+ reaches 0.937 and 0.936 with no such site. A better ranking model therefore does not remove the coverage gap. At matched 90% coverage the ensemble's sets are somewhat smaller (1.48–1.52 against 1.58–1.60 for the CNN); the paired per-site difference for LSO-CV+ is -0.059 [-0.120, $+0.002$], smaller at 6 of 8 sites, so we again call it a direction.

G. Why: validation optimism

For the CNN ERM model, we compared AUROC on seen-site validation data, the leave-site-out estimate (inner models on the sites they did not see), and the actual held-out AUROC (Fig. 5). Over 16 site-seed units, seen-site validation overstates held-out AUROC by 0.118 on average (absolute error 0.130); the leave-site-out estimate is nearly unbiased (0.007; absolute error 0.070). The pretrained head shows the same optimism (0.863 on seen-site validation vs. 0.749 on held-out sites, a gap of 0.115 over 40 units). Neither estimate ranks sites (Spearman with actual AUROC for the CNN: -0.310 and -0.048): actual AUROC ranges from 0.56 to 0.86 while the leave-site-out estimate stays between 0.72 and 0.75, so it describes a typical unseen site and not a particular one. The per-site optimism gap and pooled-conformal coverage are rank-correlated in the expected direction (-0.452 over eight sites), consistent with optimism driving under-coverage but not significant at this sample size. Averaging the inner models (a free by-product) gives AUROC 0.748 vs. 0.733 for the single model (0.016 [-0.008, 0.049]), no significant gain.

TABLE VII
CONFORMAL CALIBRATION ACROSS MODEL FAMILIES AT $\alpha = 0.1$ (EIGHT SITES, TWO SEEDS). COV. IS MEAN COVERAGE WITH, IN PARENTHESES, THE NUMBER OF SITES MORE THAN FIVE POINTS UNDER THE 0.90 TARGET; SIZE IS MEAN SET SIZE; @90% IS SET SIZE INTERPOLATED AT 90% COVERAGE OVER THE α GRID.

	CNN			Pretrained head			CNN + pretrained		
Calibration	Cov. (under)	Size	@90%	Cov. (under)	Size	@90%	Cov. (under)	Size	@90%
Pooled (seen-site val.)	0.836 (4)	1.39	1.58	0.834 (5)	1.32	1.55	0.836 (5)	1.29	1.48
Worst-site (seen-site val.)	0.904 (2)	1.60	1.58	0.890 (1)	1.52	1.57	0.889 (2)	1.48	1.52
LSO-split	0.905 (1)	1.59	1.57	0.894 (1)	1.54	1.56	0.904 (2)	1.51	1.49
LSO-CV+	0.942 (0)	1.75	1.60	0.937 (0)	1.68	1.51	0.936 (0)	1.63	1.52
LSO-worst-site	0.960 (1)	1.81	1.59	0.960 (0)	1.82	1.57	0.973 (0)	1.79	1.48

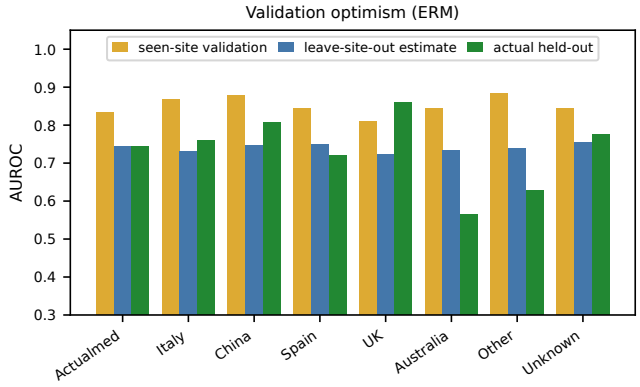


Fig. 5. AUROC of the CNN ERM model estimated on the seen-site validation data, by leave-site-out models, and obtained on the held-out site (two-seed mean).

H. Simulated shifts are poor stand-ins

On the semi-synthetic benchmark with three acquisition-severity levels, the seen-site pooled calibration already covers 0.896–0.922 at the 0.90 target, unlike the 0.836 on real sites. The LSO variants are no better, and LSO-split under-covers at the highest severity (0.834, lowest site 0.688). Therefore, the real under-coverage probably comes from differences between sites that our simulation does not contain (patient mix, case selection, labeling, and image provenance). A calibration method that was checked only on simulated acquisition shifts would have looked fine.

VI. DISCUSSION AND LIMITATIONS

A. What we think the evidence supports

On this benchmark, augmentation, site adversaries, profile adversaries, CORAL, and GroupDRO did not significantly improve the held-out AUROC, either for a small CNN trained from scratch (gains above about 0.055 are unlikely) or for heads on a frozen pretrained encoder (gains above about 0.04 are unlikely). Two directions repeated across folds and families without reaching significance: GroupDRO (about +0.02) and averaging a CNN with a pretrained head. Acquisition statistics leak the label across sites and model scores follow them within class; however, the methods that reduce this did not improve accuracy, so we cannot say that the shortcut explains the accuracy loss. Seen-site validation is optimistic, standard conformal sets under-cover, and leave-site-out calibration avoids the worst shortfalls by paying in set size; this held for two encoder families and their ensemble. We suspect that invariance fails here because the data are tiny (sites of 52–207 images), the label-correlated signal is mostly within-site, and adversaries add optimization noise that small heads cannot afford; we did not test these explanations.

B. Limitations

The study covers one modality and one task. Our environment could not reach larger or credentialed datasets and had no GPU, and we used no private data. The collection is retrospective and assembled from publications and case reports, so absolute AUROC is not clinically meaningful, and "Other" and "Unknown" mix institutions. With eight sites, tiny test sets, and no multiplicity correction, "no significant difference" is not evidence of equivalence below roughly 0.03 AUROC. The development/confirmation split was ours and not random, and tuning budgets were modest, so a better-tuned variant could still help. The pretrained encoder was frozen and not fine-tuned, and we did not check whether images in this benchmark also appear in its pretraining collections. The CNN confirmation results were known before the pretrained family was designed. No leave-site-out conformal rule has a finite-sample guarantee under site shift, and coverage is marginal over images, not conditional on class or patient. The profile estimators are proxies. Larger data, more sites, and fine-tuned encoders would be needed to show that invariance helps.

VII. CONCLUSION

An invariance recipe built from class-conditional alignment and site-profile augmentation did not beat ERM on held-out sites under a protocol that froze settings before the confirmation run, and the same held for a frozen pretrained encoder. What lasted is a label-free leak diagnostic, a measured gap between seen-site validation and leave-site-out performance, the observation that standard conformal sets under-cover unseen sites in both encoder families, the finding that a site-fold CV+ rule avoids the worst cases at a cost in set size, and the finding that simulated acquisition shifts do not reproduce the real effect. Averaging a from-scratch CNN with a pretrained model is a cheap direction worth testing on more sites.

DATA AND CODE AVAILABILITY

The benchmark is rebuilt from the three public collections by `scripts/build_covid_cxr.py`.

Code, frozen configurations, per-run results, and `scripts/final_report.py` and `scripts/round2_tables.py`, which regenerate every table, figure, and headline number, are available at https://github.com/Naveen060/scanner-institution-invariant; `experiments/ITERATIONS.md` logs every experiment, including failures. Early iteration-1 runs on two confirmation sites were never used and are excluded.

Ethics Statement

Only public, de-identified images were used, and no new data were collected from human participants.

Conflict of Interest

The authors declare no competing interests.

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The AI system Claude (Anthropic), used through the Claude Code environment, wrote the experiment code, ran the experiments, produced the tables and figures, and drafted all sections of this manuscript, including the Related Work and Discussion. The authors directed the work, edited the text for clarity and wording, and are responsible for the content. All reported numbers were generated by the released code. The chest-radiograph encoder used in the second experiment family is a model component and is cited as such.

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APPENDIX A
PER-SITE AUROC OF THE CNN METHODS

TABLE VIII
HELD-OUT AUROC PER SITE (TWO-SEED MEAN); D = DEVELOPMENT FOLD, C = CONFIRMATION FOLD.

Method	Actualmed[d]	Italy[d]	China[d]	Spain[d]	UK[c]	Australia[c]	Other[c]	Unknown[c]
ERM	0.745	0.761	0.808	0.720	0.859	0.565	0.629	0.775
GroupDRO	0.778	0.738	0.818	0.711	0.837	0.679	0.657	0.766
DANN	0.768	0.739	0.822	0.678	0.746	0.651	0.645	0.764
Generic aug.	0.785	0.757	0.769	0.680	0.828	0.625	0.619	0.792
Site-profile aug.	0.755	0.766	0.773	0.735	0.703	0.642	0.641	0.791
Profile adversary (PAI)	0.747	0.746	0.825	0.694	0.853	0.551	0.635	0.793
SIDL-cc	0.698	0.789	0.708	0.694	0.747	0.656	0.635	0.760
SIDL-phys	0.740	0.758	0.806	0.722	0.831	0.674	0.650	0.770
SIDL-full	0.763	0.777	0.743	0.667	0.782	0.415	0.626	0.837